

ANSWERS SCHEME KMJ PRA-PSPM DC014 2024/2025

Questions No	Answers	Marks
1(a)	Audio - Audio is the use of speech, music and sound effects. - Produced by vibration, as perceived by the sense of hearing. Graphic - A digital representation of non-text information Animation - Animation is a process of making a static image look like it is moving. - Animation is a series of graphical images displayed one after another to simulate movement. Text - Combination of letters that form a word or sentence that describe a topic which is known as text information. - Text is a basic element of multimedia that combine alphabets, numbers and symbol Video - Video is a series of frames containing pictures and sound played back at a fast rate. Video is a product of recording moving visual images by using a camera or video machine.	2 M 1m - element 1m - explain Select any one (1) answer

1(b)

Element	Explanation
Audio	• To capture attention: a sound in a multimedia presentation will capture an audience's attention. • To increase the associations the end-users make with the information in their mind (sound of a duck), thus increasing their retention rate. • Audio adds an exciting dimension to a presentation.
Graphic	• Convey information more quickly than when using text. • Make complex information easy to understand. • To present information more attractively. • To illustrate ideas through still pictures. • Images - used to add emphasis, direct attention, illustrate concepts and provide background content. • Complementary element - used as background and texture.
Animation	• Animation is useful for illustrating concepts which involve movement. • Can help organize thoughts and illustrate processes. • Can graphically simplify complicated concepts and convey complex inter-relationships, which are difficult to visualize. • Inject humor and light-heartedness in a presentation via an animated cartoon. • Can attract audience attention
Text	• Effective for conveying basic/fundamental information
Graphic	• Videos engage students of all ages and abilities • Videos motivate with authentic content • Videos provide concept/context/real situation to help understanding.

2 M

1m - keyword

1m - explain

**Select any
one (1)
answer**

1(c)	Limited detail Vector graphics are not as effective for complex images, such as photographs, that require color shading, blending, shadows, and highlights. Time and skill Vector graphics can take more time and skill to create because each vector needs to be placed manually. Compatibility Vector graphics may not be supported by all image formats and software. Some web browsers and applications may not display vector graphics correctly or at all. Inconsistent Vector images can vary from one application to another. Conversion It's generally more difficult to convert a raster image to a vector file than the other way around. Software You need a vector-based design program like Adobe Illustrator to open and work on vector files.	2 M 1m - keyword 1m - explain Select any one (1) answer										
2(a)	<table><tr><th>Scenario</th><th>Phase</th></tr><tr><td>(i) The content of a multimedia contents will be delivered online, and the students will access it on their smartphones, tablets, and laptops.</td><td>Post-production</td></tr><tr><td>(ii) You are hired to shoot and edit a highlight video for a corporate event. The video needs to showcase key moments from a conference and be ready for publishing online within 48 hours.</td><td>Production</td></tr><tr><td>(iii) The budget for the entire multimedia project is RM10,000, and the company has a tight timeline of 4 weeks to complete the production. You need to ensure the production efficiently to stay within budget and meet the deadline.</td><td>Pre-production</td></tr><tr><td>(iv) A multimedia e-commerce website is to be developed for a new fashion brand for a specific target audience, and the site should reflect a modern, user-friendly design.</td><td>Pre-production</td></tr></table>	Scenario	Phase	(i) The content of a multimedia contents will be delivered online, and the students will access it on their smartphones, tablets, and laptops.	Post-production	(ii) You are hired to shoot and edit a highlight video for a corporate event. The video needs to showcase key moments from a conference and be ready for publishing online within 48 hours.	Production	(iii) The budget for the entire multimedia project is RM10,000, and the company has a tight timeline of 4 weeks to complete the production. You need to ensure the production efficiently to stay within budget and meet the deadline.	Pre-production	(iv) A multimedia e-commerce website is to be developed for a new fashion brand for a specific target audience, and the site should reflect a modern, user-friendly design.	Pre-production	6 M
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	(v) Part of the activities in multimedia development is to listen particularly to the vocal level in the audio used and the consistency of the sound between tracks.	Post-production	
	(vi) The script of a video demonstration project has been written by the marketing team and includes voiceover text, on-screen text, and a call-to-action. You are to ensure the script is clear, concise, and free of any errors or inconsistencies before it moves into the production phase.	Production	
2 (b) i)	Description	Type authoring tool	2 M
	Best used when the bulk of your content consists of elements that can be viewed individually	Card based	
	These tools are best to use for those projects, wherein the information flow can be directed from beginning to end much like the movies.	Time based	
2(b)(ii)	Icon based authoring tool		1 M
2(b)(iii)	• <i>Difficult to learn</i> as user may struggle to navigate the tool's advanced features of the tool • Professional or enterprise versions can be <i>more expensive</i>, with pricing often based on subscription or licensing fees		2 M <i>Choose 1 Any acceptable answer</i>
2(c)(i)	Offline medium delivery		1 M

2(c) ii)	• CD-ROM • DVD-ROM • Blu rays • USB flash drives • External hard drives • Printed materials with digital content • Interactive Kiosks	2 M <i>Choose 2 Any acceptable answer</i>															
2(c)(iii)	• Limited content updates: Once the content is distributed on physical media, it cannot be updated without redistributing new versions, making it difficult to keep content current. • Storage limitations: Physical media has limited storage capacity, which can restrict the amount of multimedia content that can be delivered, especially for large files like high-definition videos or complex software • Physical damage: Media such as CDs, DVDs, or USB drives can be easily damaged, lost, or corrupted, potentially making the content inaccessible.	2 M <i>Choose 1 Any acceptable answer</i>															
3(i)	<table><tr><td></td><td>File Processing System</td><td></td><td>Database System</td><td></td></tr><tr><td>Data Redundancy</td><td>Data is often duplicated / High data redundancy. [J1]</td><td>Minimize data redundancy / Low data redundancy. [J1]</td><td></td><td></td></tr><tr><td>Data Sharing</td><td>Limited data sharing as data is stored in separate files. [J1]</td><td>Facilitates efficient data sharing among multiple users. [J1]</td><td></td><td></td></tr></table>		File Processing System		Database System		Data Redundancy	Data is often duplicated / High data redundancy. [J1]	Minimize data redundancy / Low data redundancy. [J1]			Data Sharing	Limited data sharing as data is stored in separate files. [J1]	Facilitates efficient data sharing among multiple users. [J1]			4 M
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3(ii)	Reduce data redundancy <i>With a database there is no need to repeat recording the same data.</i> Improve data integrity <i>Ensures that data is correct for all files</i> Improve data security <i>Data can be managed by authorize user only</i>	2 M <i>Select any one (1) answer</i>															
4 (i)	A condition in which the data in the database complies with all entity and referential integrity constraints. All data in the database is consistent with real-world events and conditions	2 M															

4 (ii)	Description	Database Object	2 M
	Design a structured format that stores each product's details, including ID, name, price, and stock quantity	Table	
	Generate a printable document that shows the total sales and revenue generated each month.	Report	
5 (a)	A software used to monitor the performance and health of computer applications /network systems in real-time so that they run smoothly.		2 M

5(b)	Telemedicine/Videoconferencing/collaborative computing/telecommuting/Multimedia networking/Television and radio broadcasting		2 M
5(c)	A user requires a stable and fast internet connection at home from the internet service provider to support multiple devices for streaming, gaming, and remote work.	Optical fiber	4 M
	A company needs to set up a local area network (LAN) to enable efficient communication and data transfer between various departments in a multi-story office building.	Twisted pair cable	
	A small business setting up a security camera system throughout its office. They need reliable, interference-resistant connections to ensure that the video feeds from the cameras are clear and uninterrupted to carry high-frequency signals over longer distances without significant degradation.	Coaxial cable	
	A major hospital needs a high-speed, high-capacity network to support its vast array of medical devices, patient records, and communication systems. It must be able to transmit data at incredibly high speeds, and immune to electromagnetic interference.	Optical fiber	

5(d)	Characteristics	Type of wireless transmission media	4 M
	Base stations will serve a region where it will transmit and receive high frequency radio waves, facilitating communication between your device and the broader network.	Cellular radio	
	Transmission that uses high-frequency radio waves to transmit data over long distances. These signals can travel through the atmosphere and are commonly used for point-to-point communication, such as between cellular towers or for satellite uplinks.	Microwave	
	Involves sending signals from a ground station to a satellite in orbit. The satellite then relays these signals back to another ground station, allowing for communication over vast distances	Satellite	
	Transmissions work over short distances and require a line of sight between the sending and receiving devices. Think of it as an invisible light beam that carries your data, allowing devices to communicate without cables.	Infrared	
5(e)	Function	Communication device	6 M
	This device is translating data from your computer into signals that can travel over the network and vice versa that is usually attached to the computer.	Network Interface Card	1m each
	It directs data between your home or business network and the wider internet, ensuring information gets to the right destination.	Router	
	Device that translates digital data from your computer into analog signals that can travel over telephone lines, cable systems, or satellite connections—and vice versa.	Modem	
	A signal boost for your network. Its job is to take weaker or fading signals and amplify them	Repeater	

	so they can travel longer distances without losing strength.			
	A bridge between your wired network and your wireless devices. It allows Wi-Fi devices like phones, laptops, and tablets to connect to your network without needing physical cables.		Wireless Access Point	
	A device that links multiple devices on a network, allowing them to communicate directly and efficiently with each other.		Switch	
6 a) i)	search engine		1M	
6 a) ii)	A software system designed to locate web pages about specific topics by key in a word or phrase, called keyword		2 M	
6(b)	Description	Type of search strategies	3M	
	You are looking for information on Java programming, but you want to exclude results that are related to coffee.	boolean		
	Finding the article on internet of thing in Journal of big Data.	Field		
	Using quotation marks around the phrase ensures that the search engine looks for the exact phrase "internet of things" rather than individual words.	phrase		
6(c)				
	Scenario	web activities	Example	8 M
	Lisa always chat her mother whenever she miss her	Online messaging	Whatapp, telegram	
	Khairul amin manage to all his "sambal Podeh" in five minutes after he launch the product online.	Shopping online	Shoppe, tik tok shop	
	Reena share her personal content, interests, activities, backgrounds or real-life connections.	Social Network	Instagram/ Facebook/Tik tok	
	Iman completed the 2 chapters online course in order to get a certificate.	Online education	Ibmskillbuild /Coursera, Udemy/edX/ Khan Academy	

6(d) i)	• Through cookies • Spyware installation • Virus infections * <i>Choose 1</i> * <i>Any acceptable answer</i>		1M								
6(d) ii)	<table><tr><td>Through cookies</td><td> • Periodically clearing her cookies history to ensure that trackers and advertising companies can't retain data about her online behaviour • She can install browser extensions or add-ons that block cookies where it will block or manage cookies more effectively and can automatically delete cookies when she closes her devices </td><td rowspan="4">2M</td></tr><tr><td>Spyware installation</td><td> • Install antispyware programs that can detect and remove spyware from her devices. She needs to make sure to keep the software up to date to catch the latest threats • She can enable a firewall to block unauthorized connections and monitor traffic to and from the device preventing spyware from communicating with external servers </td></tr><tr><td>Virus infections</td><td> • Install and enable antivirus software that provides real-time protection against malware and viruses. This will monitor web browsing activities and block malicious websites or downloads that may attempt to infect your device • Disable any settings in the browser that automatically download files. This prevents any potentially harmful files from being automatically saved to the devices without consent </td></tr><tr><td colspan="2"> * <i>Choose 1 with correct label from the answer in question 6(d)(ii)</i> * <i>Any acceptable answer</i> </td></tr></table>	Through cookies	• Periodically clearing her cookies history to ensure that trackers and advertising companies can't retain data about her online behaviour • She can install browser extensions or add-ons that block cookies where it will block or manage cookies more effectively and can automatically delete cookies when she closes her devices	2M	Spyware installation	• Install antispyware programs that can detect and remove spyware from her devices. She needs to make sure to keep the software up to date to catch the latest threats • She can enable a firewall to block unauthorized connections and monitor traffic to and from the device preventing spyware from communicating with external servers	Virus infections	• Install and enable antivirus software that provides real-time protection against malware and viruses. This will monitor web browsing activities and block malicious websites or downloads that may attempt to infect your device • Disable any settings in the browser that automatically download files. This prevents any potentially harmful files from being automatically saved to the devices without consent	* <i>Choose 1 with correct label from the answer in question 6(d)(ii)</i> * <i>Any acceptable answer</i>		
Through cookies	• Periodically clearing her cookies history to ensure that trackers and advertising companies can't retain data about her online behaviour • She can install browser extensions or add-ons that block cookies where it will block or manage cookies more effectively and can automatically delete cookies when she closes her devices	2M									
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Virus infections	• Install and enable antivirus software that provides real-time protection against malware and viruses. This will monitor web browsing activities and block malicious websites or downloads that may attempt to infect your device • Disable any settings in the browser that automatically download files. This prevents any potentially harmful files from being automatically saved to the devices without consent										
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6 d) iii)	• Violation of Free Speech • Lack of Transparency • Potential for authoritarian threats and abuses of government power • Restriction to an important information	1M Select any one (1) answer
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6 e)	• Ensuring data security is a major challenge as IoT devices collect vast amounts of personal and sensitive data. Unauthorized access or data breaches can expose individuals and organizations to cyberattacks, identity theft, or data manipulation. • IoT devices require constant, high-quality internet connectivity, but in many regions, the necessary infrastructure is underdeveloped, limiting their effectiveness in the 4IR • Integrating IoT within the 4IR requires skilled professionals and organizations adaptable to new technologies, alongside investment in training and fostering innovation and collaboration across industries	2M - <i>Select any one (1) answer</i>
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